Dashboard Reporting and Application Reference

American Forests Carbon Dashboard

Spatial Informatics Group

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Dashboard Workflow Overview

The dashboard uses three main pages for navigating the dashboard:

- Project Builder
- Frequently Asked Questions
- Model Management (Visible only when ‘ENV’ is not production)

In production, Model Management is hidden.

The dashboard supports two primary analysis paths. In Project Builder, Planting Design is used when the user has a defined project acreage and wants to evaluate carbon and financial performance. The Solver is used when the user starts with a financial target or wants to evaluate breakeven acreage or planting density.

Primary Planting Design Flow

Select Site	Define Planting Design	Review Carbon Estimates	Review Project Financials	Export Results

Solver Breakeven Analysis Flow

Select Site	Configure Scenario	Set Financial Target	Run Solver	Review Breakeven Result

Both analysis paths begin with Site Selection because the selected FVS variant/location determines the model options, species configuration, PCT options, and other downstream inputs available to the user.

Project Builder User Flows

The workflows below describe the major user actions and decision points for various evaluations and analysis that can be run with the dashboard.

The left sidebar shows the Project Workflow list from *conf/base/workflow_steps.json*. The active workflow step is highlighted.

Fixed Acreage Project Evaluation

Site Selection

Use the Site Selection Map to choose the FVS variant location geometry for the project area. The selected location controls the species options, model defaults, PCT options, and downstream calculations.

The user begins by locating the project through latitude/longitude, address entry, or an uploaded project geometry such as GeoJSON, Shapefile, or a zipped file with geometry components.

There are two options for uploading a file. The first (top) option is Streamlit's default file upload widget. This option is quicker but has file size restraints that can possibly limit the user from uploading a file. If this occurs, option two (bottom) is available. This option allows for larger file uploads as the files are uploaded to a Google Cloud Storage (GCS) bucket and read from their GCS URL. The bucket has a retention period for uploaded files so that they're automatically deleted after a certain period of time.

The dashboard attempts to auto-select a supported FVS variant/location when a point, address, or uploaded geometry intersects a supported map feature. If the geometry intersects multiple variants, the user will need to select which variant they would like to be used.

At the bottom of Site Selection, the Planting Design and Solver buttons are disabled until a supported location is selected.

Planting Design

Use the Planting Design to estimate carbon units and financial results for a fixed project acreage. This page is organized into three main expandable sections:

- Planting Parameters
- Carbon Estimates, and
- Project Financials.

Planting Parameters

After a supported location is selected, the user configures acreage, survival percentage, site index where applicable, species trees/acre, pre-commercial thinning, and other related planting

inputs. If the selected map variant has multiple model variants or sub-variants available for the selected location, the page resolves those options and uses the active variant for downstream calculations. These inputs control the model run and drive the carbon chart shown beside the controls.

Net acres is the project area used to scale per-acre carbon and credit values up to total project values. Survival percentage represents the expected survival rate for planted trees. Site index represents site productivity; some variants may not use site index, in which case the dashboard handles that input differently.

The species mix inputs correspond to the species in the selected variant's configured species order. The dashboard also calculates total TPA across all selected species. If total TPA exceeds the configured cap, the dashboard warns that the scenario may be unrealistic or outside the intended modeling range.

The PCT selector represents pre-commercial thinning. It is based on registry data for the selected variant location. Typical levels are PCT0, PCT1, and PCT2. When retention percentages are available, they are displayed with the PCT options.

The charts in this expander update from the planting inputs. They show the projected carbon-related model output for the selected planting design. If these charts do not update, check that a location is selected and that at least one species has a TPA value greater than zero.

Carbon Estimates

The Carbon Estimates expander converts the planting scenario into carbon and carbon-unit outputs. It depends on the Planting Parameters section since it uses the current variant, location, PCT level, survival percentage, site index, acreage, and species TPA values.

The dashboard first checks that carbon data exists from the planting inputs. If no carbon data exists, the user should return to Planting Parameters and adjust the sliders or required inputs.

This section includes a CO2e accumulation summary. The summary interpolates modeled carbon accumulation and reports values at major project horizons such as 10, 50, and 100 years. These values are shown in tons CO2e and help summarize long-term project climate benefits.

The user selects one or more carbon protocols. Supported protocol names include ACR, CAR, VERRA, GS, and ISO. The selected protocols control which carbon-unit lines and tables are generated.

Behind the scenes, the dashboard sends carbon rows and protocol selections to the model service. The service applies protocol rules such as conversion coefficients and buffer deductions where applicable. The resulting carbon-unit data is stored for use in the Project

Financials section.

This section may also provide controls for viewing values as total project acreage or per-acre values, depending on the chart being displayed. Total project values scale by net acres, while per-acre values show intensity independent of project size.

If carbon estimates are empty, confirm that a location is selected, species TPA values are populated, protocols are selected, and the model service is reachable.

Project Financials

The Project Financials expander converts carbon units into financial results. It uses the selected protocols and generated carbon-unit rows from Carbon Estimates.

Some assumptions are shared across protocols, while some can be edited by protocol where the dashboard allows protocol-specific values. Net acres flow from Planting Parameters and are used to scale project costs and revenues.

Financial assumptions include editable inputs:

- Initial planting cost per acre
- Initial credit price

Financial assumptions include non - editable inputs:

- Number of plots
- Cost per CFI plot
- Registry fees
- Issuance fee per credit
- Validation cost
- Verification cost
- Anticipated inflation
- Discount rate
- Credit price increase

The NPV Year Horizon selector controls how many years from the project start are included in the discounted cashflow calculation. Available horizons are shown as fixed year options.

The model service calculates proforma rows for each selected protocol. The proforma includes revenues, costs, and net revenue over time. Credit sales are modeled on a recurring schedule, and costs can include planting, validation, verification, survey or plot costs, registry fees, and issuance fees.

The net revenue chart shows financial performance over time. A display toggle can switch between total project values and per-acre values. Total values reflect the selected net acreage,

while per-acre values divide the financial result by project acres.

The Project Financials Summary table compares protocols using metrics such as total net revenue, NPV at the selected horizon, and NPV per acre. This table is useful for comparing how protocol choice and assumptions affect financial feasibility.

The user can download the proforma table as a CSV. This export contains the detailed annual rows used to create the financial chart and summary table.

After financial results are generated, the user can select Generate Project Report. This sends the planting design, species mix, financial assumptions, carbon results, and selected variant to the model service. The service generates a PDF report using Quarto, and the dashboard provides a PDF download when generation succeeds.

Recommended workflow for this page:

1. Complete Planting Parameters.
2. Review Carbon Estimates and select protocols.
3. Review Project Financials and adjust assumptions.
4. Download the proforma CSV, if needed.
5. Generate and download the project report, if needed.

Solver Breakeven Evaluation

Use the Solver when the project is being evaluated against a financial target rather than starting with a known project acreage. The Solver uses the selected planting scenario, carbon protocol, and financial assumptions to estimate the acreage or planting density required to meet the specified financial target.

The Solver supports two primary analysis paths:

- Breakeven Acreage Analysis – estimates the net project acreage required to reach a target NPV.
- Planting Density Breakeven Analysis – evaluates the planting density required to break even for a fixed project acreage.

Site Selection

The Solver begins with the same Site Selection process used for a fixed-acreage project evaluation. Use the Site Selection Map to choose the supported FVS variant location geometry for the project area.

The user can locate the project using latitude/longitude, an address, or an uploaded project geometry. The selected location determines the model options, species configuration, PCT options, and other inputs available for the Solver analysis.

The Solver cannot be used until a supported project location has been selected.

Configure the Planting Scenario

After selecting a project location, configure the planting scenario that will be evaluated by the Solver.

The planting scenario defines the biological and modeling assumptions used to generate the underlying carbon estimates. These include the applicable model variant, species mix, trees per acre (TPA), survival percentage, site index where applicable, pre-commercial thinning (PCT), and other planting inputs available for the selected location.

For a breakeven acreage analysis, these planting assumptions are held constant while the Solver determines the project acreage required to meet the financial target.

For a planting density analysis, project acreage is fixed and planting density becomes the variable evaluated by the Solver.

Select Carbon Protocol

Select the carbon protocol to use for the Solver analysis. The selected protocol determines the carbon-unit rules applied to the modeled carbon estimates and provides the carbon-unit values used in the financial calculations.

The solver evaluates the planting scenario using the assumptions associated with the selected protocol so that the resulting breakeven value reflects both modeled carbon performance and protocol-specific carbon-unit calculations.

Configure Financial Assumptions

Set the financial assumptions used to evaluate the scenario. These assumptions determine the costs, revenues, and discounted financial performance associated with the modeled project.

The Solver uses these assumptions together with the carbon-unit results to evaluate project financial performance. Changes to assumptions such as planting costs or credit prices can therefore change the acreage or planting density required to reach the financial target.

This allows the Solver to be used not only to identify a single breakeven result, but also to examine how the result changes under different financial assumptions.

Set the Financial Target

For a breakeven acreage analysis, enter the target NPV that the project should achieve.

Unlike the fixed-acreage workflow, acreage is not entered as the primary project constraint. Instead, the Solver evaluates the configured planting and financial scenario and determines the estimated net acreage required to reach the specified NPV target.

The analysis therefore asks, given this planting design, carbon protocol, and set of financial assumptions, how many net acres are required to reach the target NPV?

Run the Breakeven Acreage Solver

Run the Solver after the planting scenario, protocol, financial assumptions, and target NPV have been configured.

The Solver evaluates the scenario through the model service and searches for the net acreage at which the modeled project reaches the specified financial target.

The resulting breakeven acreage represents the estimated project size required for the configured scenario to meet the target NPV. If the planting design, protocol, financial assumptions, or target changes, the Solver should be run again to evaluate the updated scenario.

Review the Solver result to determine the estimated net acreage required for the project to reach the specified target.

The result should be interpreted in the context of the assumptions used to generate it. For example, changing planting costs, credit prices, species configuration, planting density, or protocol assumptions may change the acreage required to reach the same financial target.

The Solver can therefore be used iteratively to compare how different project assumptions affect the scale required for financial feasibility.

Planting Density Breakeven Analysis

The Solver can also evaluate planting density when project acreage is already known. In this workflow, acreage is fixed and the Solver varies TPA to identify the planting density associated with breakeven.

The user first selects the project location, defines the fixed acreage, configures the species mix, selects the carbon protocol, and sets the financial assumptions. The user then chooses how planting density should be varied.

Three density sweep modes are available:

- Scalar – scales the entire species mix while preserving the relative proportion of species.
- Species – varies one selected species while holding the other species constant.
- Per-species – evaluates each species separately.

The Solver evaluates the selected density range and returns the TPA result associated with the breakeven condition. This allows the user to examine how planting density and species configuration affect project financial performance.

Recommended Breakeven Acreage Workflow

1. Select a supported project location.
2. Configure the planting scenario.
3. Select the carbon protocol.
4. Configure the financial assumptions.
5. Enter the target NPV.
6. Run the Solver.
7. Review the estimated breakeven acreage.
8. Adjust planting or financial assumptions and rerun the Solver as needed.

Recommended Planting Density Workflow

1. Select a supported project location.
2. Define the fixed project acreage.
3. Configure the species mix and planting assumptions.
4. Select the carbon protocol.
5. Configure the financial assumptions.
6. Choose the density sweep mode.
7. Run the Solver.
8. Review the resulting TPA values and compare scenarios as needed.

Supporting and Administrative Functions

Model Management

Model Management is available only outside production. It is intended for administrators and developers managing model availability.

Users can upload their models using the Upload Model tab. Users have the capabilities to bulk upload regression model *.pkl* files.

The upload helper expects filenames shaped like:

{Variant}_v{Version}_{PCT}_{Method}_{LocCode}_ridge_models.pkl

Example:

PN_v1_PCT0_Jenkins_609_ridge_models.pkl

After a model is uploaded, the application automatically validates the model file structure and extracts metadata from the filename. It then assigns the appropriate variant, location code, PCT level, retention percentage, and version information. Administrators can configure species-specific settings, including species order and default trees-per-acre (TPA) values, and define maximum TPA caps before the model is made available for use.

The Model Registry provides a centralized inventory of all models currently available in the application. Administrators and developers can browse the registry to review model metadata, verify that models have been successfully registered, and confirm the available variants, location codes, PCT levels, retention percentages, versions, and species configurations. This interface provides a quick overview of the current model catalog without requiring access to the underlying model files.

Glossary

The Glossary provides clear definitions for important terms used throughout the dashboard, including forestry metrics, carbon accounting concepts, protocol names, financial inputs, and solver outputs. It is designed to help users quickly understand dashboard terminology, common abbreviations, and where each concept appears in the codebase, with searchable entries grouped by category.

Frequently Asked Questions

The FAQ page provides reference information on baseline assumptions, protocol differences, carbon accounting assumptions, FVS and machine-learning approximation concepts, unrealistic scenario warnings, and verification cost logic.